The Effect of Multi-Ingredient Protein versus Collagen Supplementation on Satellite Cell Properties in Males and Females

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ABSTRACT

WAGEH, M., S. A. FORTINO, R. PONTELLO, A. MAKLAD, C. MCGLORY, D. KUMBHARE, S. M. PHILLIPS, and G. PARISE. The Effect of Multi-Ingredient Protein versus Collagen Supplementation on Satellite Cell Properties in Males and Females. *Med. Sci. Sports Exerc.*, Vol. 56, No. 11, pp. 2125–2134, 2024. **Introduction:** Skeletal muscle satellite cells (SC) contribute to the adaptive process of resistance exercise training (RET) and may be influenced by nutritional supplementation. However, little research exists on the impact of multi-ingredient supplementation on the SC response to RET. **Purpose:** We tested the effect of a multi-ingredient supplement (MIS) including whey protein, creatine, leucine, calcium citrate, and vitamin D on SC content and activity as well as myonuclear accretion, SC and myonuclear domain compared with a collagen control (COL) throughout a 10-wk RET program. **Methods:** Twenty-six participants underwent a 10-wk linear RET program while consuming either the MIS or COL supplement twice daily. Muscle biopsies were taken from the vastus lateralis at baseline and 48 h after a bout of damaging exercise, before and after RET. Muscle tissue was analyzed for SC and myonuclear content, domain, acute SC activation, and fiber cross-sectional area (fCSA). **Results:** MIS resulted in a greater increase in type II fCSA following 10 wk of RET (effect size (ES) = 0.89) but not myonuclear accretion or SC content. Change in myonuclei per fiber was positively correlated with type I and II and total fiber hypertrophy in the COL group only, indicating a robust independent effect of MIS on fCSA. Myonuclear domain increased similarly in both groups, whereas SC domain remained unchanged following RET. SC activation was similar between groups for all fiber types in the untrained state but showed a trend toward greater increases with MIS after RET (ES = 0.70). **Conclusions:** SC responses to acute damaging exercise and long-term RET are predominantly similar in MIS and COL groups. However, MIS can induce greater increases in type II fCSA with RET and potentially SC activation following damage in the trained state. **Key Words:** SATELLITE CELLS, WHEY PROTEIN, LEUCINE, CREATINE, EXERCISE, MUSCLE DAMAGE

Resistance exercise training (RET) is a known stimulus for inducing skeletal muscle hypertrophy (1). Satellite cells (SC) play an important role in the repair process following damage and are associated with muscle hypertrophy in humans following RET (2,3). Hypertrophy is induced following repeated exposures to load-bearing stimuli, which could lead to damage requiring repair and adaptation. Following damage, quiescent SC activate, enter the cell cycle, and proliferate before terminally differentiating and fusing with existing fibers, or returning to the quiescent state to repopulate the basal SC pool (3,4). This process is orchestrated by the upregulation of the paired box transcription factor 7 (Pax7) along with myogenic regulatory factors MyoD, Myf5, MRF4, and Myogenin (3,5).

Resistance exercise is known to promote skeletal muscle hypertrophy, which can be augmented by the addition of nutritional supplementation (6–8). Recent work by our group (9) and others (10–12) demonstrated that supplementation (with whey protein alone or in a multi-ingredient format) can augment skeletal muscle hypertrophy in conjunction with exercise training. Whey protein is a potent anabolic stimulus because of its high essential amino acid content and, in particular, the presence of leucine, which is known to stimulate muscle protein synthesis through mammalian target of rapamycin signaling (13). Other supplements, such as creatine, calcium citrate, and vitamin D, are also known to increase lean tissue mass and muscle strength with resistance training longer than 6 wk in both young and old populations (14–16).

There is evidence suggesting that nutritional supplementation may augment SC function, amplifying the hypertrophic response as well as exercise performance (17–20). One of the first reports of oral nutritional supplementation to impact

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SC appeared in 1999, reporting that creatine, combined with an overload stimulus, was able to enhance SC mitotic activity in rodents (17). This was confirmed in humans several years later through a 16-wk training study in young men, where creatine supplementation augmented the RET-induced increase in SC number and myonuclear concentration in skeletal muscle fibers (19). Similarly, whey protein may accelerate the proliferation of type II fiber-associated SC in response to an acute bout of exercise (18). Leucine plays a key role in regulating cell growth and protein synthesis, as seen both *in vitro* and *in vivo* (21–23). Vitamin D supplementation has been implicated in influencing SC activity and function in response to damaging exercise through interaction with the vitamin D receptor found on SC (24,25). Vitamin D is also recognized for its role in calcium and phosphate absorption and homeostasis, and thus is often studied in conjunction with calcium supplementation. The combination of vitamin D and calcium has been shown to exert beneficial effects on SC activity and fiber cross-sectional area (fCSA) (26,27).

It is currently unknown whether a multi-ingredient supplement (MIS) can affect SC properties, especially in conjunction with RET. It can be speculated that the effect of supplementation on SC function would be in alignment with the impact of supplementation on skeletal muscle hypertrophy, because increases in muscle fiber size are often supported by an increase in SC content (17,28,29). Therefore, the purpose of the present study was to investigate the effects of a MIS on SC activity against a collagen control (COL) in healthy young males and females in conjunction with a 10-wk linear RET program. The current report is a secondary analysis on a previously published study (9). We also sought to assess the SC response following acute damage in untrained versus trained individuals. We hypothesized that MIS would not mitigate the response to acute damaging exercise after 48 h, but it would elicit greater increases in SC expansion, activation, myonuclear accretion, and myonuclear domain following training as compared with COL supplementation.

METHODS

Participants. Twenty-seven healthy young adults (13 male, 14 female) defined as recreationally active (i.e., partaking in nonresistance exercise no more than twice per week) were recruited for this study. This study was conducted as part of a larger trial investigating supplementation effects with RET on skeletal muscle hypertrophy (9). All participants were screened and informed of the purpose, experimental procedures, and risks of the study before providing written consent. Participants were nonsmokers, were free of any acute or chronic illness, and did not take any prescription or nonprescription medication (except for birth control; 12 of the female subjects were normally menstruating, and 2 using second-generation oral contraceptives). Participants were not consuming any nutritional supplement or medication known to affect protein metabolism (i.e., corticosteroids, nonsteroidal anti-inflammatory drugs, protein, or creatine). All participants were deemed eligible for exercise according to the Get Active Questionnaire (https://store.csep.ca/pages/getactivequestionnaire). One subject (female) dropped out as a result of injury sustained in a non–intervention-related event, leaving $n = 13$ per group. This study was approved by the Hamilton Health Sciences Integrated Research Ethics Board (Reference No. 4449) and was registered at https://clinicaltrials.gov/NCT03525197. This protocol adhered to the Declaration of Helsinki on research involving human subjects.

Study design. This was a randomized, double-blind controlled study. Participants were randomized into either the active supplementation group (whey protein isolate (20 g; 895 WPI, Fonterra), creatine monohydrate (2.5 g; Creapure, Alzchem), leucine (2 g; Ajinomoto), calcium citrate (300 mg), and vitamin D (1000 IU; MIS; $n = 13$)) or an isonitrogenous collagen control group (collagen peptides (20 g, BodyBalance, Gelita), alanine (1.4 g, Ajinimoto), glycine (0.6 g, Ajinimoto); COL; $n = 13$). Supplements were matched for flavor (citrus) as well as smell and consistency in the original powdered formula. Supplements were taken twice daily in conjunction with a 10-wk linear resistance training program. Initial assessments consisted of a body composition scan via dual-energy x-ray absorptiometry and anthropometric measurements (height, mass). Shortly after, the first muscle biopsy was taken from the vastus lateralis. Immediately after, participants underwent an acute bout of 300 maximal eccentric leg extensions (30 sets × 10 reps, $180°\cdot s^{-1}$, 60-s rest) against a Biodex Dynamometer (Biodex-System 3; Biodex Medical Systems, Inc., Shirley, NY), and another biopsy was taken 48 h later. After 10 wk of the resistance training program, another muscle biopsy was taken from the same leg (at least 72 h after the last resistance training bout), followed by the same protocol of 300 eccentric leg extensions and a biopsy 48 h later.

Resistance training intervention. Participants underwent a linear resistance training program, with a progressive increase in load (~5%) between weeks to ensure an overload stimulus and training to failure. Each week consisted of 4 d of training, with alternating days focused on the lower body (Monday + Thursday: squats, deadlifts, lunges, hamstring curl, leg press, calf raise, leg extensions) and the remaining days focused on the upper body (Tuesday + Friday: bench press, standing military barbell press, incline dumbbell press, skull crushers and assisted dips, close-grip barbell press, single-arm row, seated row, pronated and supinated lat pull-downs, supinated incline bicep curls, planks and sit-ups). The full RET protocol is outlined in Table 1 of Ref. (9).

Muscle biopsy sampling. Skeletal muscle biopsies were taken from the vastus lateralis at baseline and following damage before and after 10 wk of resistance training. Participants arrived fasted to the testing facility and had refrained from exercise for at least 72 h. Muscle biopsies were performed under local anesthesia (2% xylocaine) using a 5-mm Bergstrom needle adapted for manual suction. Approximately 150 mg of muscle tissue was collected from each biopsy, and the tissue was cleared of any connective or adipose tissue and oriented longitudinally in optimal cutting temperature (OCT) compound

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(Tissue-Tek; Sakura Finetek, Torrance, CA). Samples were frozen in isopentane cooled using liquid nitrogen and stored in −80°C until analysis. Approximately 40 g of tissue was sectioned for immunohistochemical analysis.

Immunohistochemistry. Immunofluorescent staining was used to assess fiber type and fCSA, and enumerate SC content and activation status in each sample cut using Pax7$^+$ (canonical SC marker) and MyoD$^+$ (myogenic regulatory factor used to classify SC activation). SC activation was assessed before training (untrained state response (UTSR)) and after training (TSR; trained state response). Muscle cross sections were sectioned (5 μm) at −20°C using a cryostat microtome (Microm HM550 Cryostat; Thermo Fisher Scientific, Waltham, MA). Sections were mounted on glass slides and stored at −80°C until subsequent analysis. Slides were stained with antibodies against Pax7 (neat; Developmental Studies Hybridoma Bank), MyoD (anti-MyoD1; clone 5.8A, 1:100; Dako), A4.951 (myosin heavy chain (MHC) type I, slow isoform, 1:1; Developmental Studies Hybridoma Bank), and MHC-II (fast isoform, 1:1000; ab91506; Abcam). Secondary antibodies used were Pax7 (Alexa Fluor 488 or 594, 1:500; Invitrogen, Molecular Probes), MyoD (biotinylated secondary antibody, 1:200 (Vector Laboratories); and streptavidin-594 fluorochrome, 1:200 (Invitrogen, Molecular Probes)), A4.951 (Alexa Fluor 488, 1:500), MHC-II (Alexa Fluor 647, 1:500), and laminin (Wheat Germ Agglutinin 488 or 647, 1:200; Vector Laboratories). Myonuclei were labeled with 4,6-diamidino-2-phenylindole (DAPI; 1:20,000; Sigma-Aldrich) followed by the mounting of a coverslip with fluorescent mounting media (Dako). Slides were imaged with a Nikon Eclipse Ti Microscope equipped with a Photometrics CoolSNAP HQ2 fluorescent camera (Nikon Instruments). Images were captured and analyzed using Nikon NIS Elements software (AR 3.2, Nikon Instruments). All images were captured using the 20× objective, with approximately 250 fibers included per participant per time point in the analyses for muscle fCSA, SC content, activation, and myonuclear domain and accretion. fCSA was quantified by encircling all fibers within an image and calculating the average area of these fibers. Myonuclear domain was quantified as the proportion of average fiber size by the average number of myonuclei per fiber (30).

Statistical analyses. An *a priori* sample size estimation was conducted using G*Power (Version 3.1.9.4, Kiel, Germany) for the primary analysis of skeletal muscle hypertrophy, giving a final sample size requirement of 26 subjects. Independent Student's *t*-tests were used to assess potential differences at baseline between groups. SC content and activation data were analyzed using a two-way repeated-measures ANOVA with treatment (MIS vs COL) and time (pre and post) as the two factors. Where significance was found, a Tukey's *post hoc* test was performed. For all analyses, statistical significance was set to $P < 0.05$. Values are presented as means ± SEM. Effect sizes (ES) are reported as Cohen's *d* to infer the magnitude of differences in change scores between groups and are classified as small (0.2), moderate (0.5), and large (0.8) according to Cohen (31).

RESULTS

Participant Characteristics

Twenty-six healthy young adults completed the study. Complete subject characteristics are reported in Ref. (9). There were no significant differences between groups at baseline in body composition, strength, or dietary intake ($P < 0.05$).

Body Composition and Maximal Strength

As reported previously (9), both groups significantly increased total LBM with training (as measured via dual-energy x-ray absorptiometry), with a greater increase seen in the MIS group ($P < 0.05$; ES = 0.82). No differences were observed between groups over time for FM, %BF, BMC, or BMD ($P > 0.05$). Similar increases in strength were observed between groups in bench press, leg press, lat pull-down, and MVC torque ($P > 0.05$). All body composition and strength data are reported in (9).

Skeletal Muscle fiber Characteristics

Myofiber CSA was significantly greater after 10 wk of RET in both groups, with no significant differences observed between groups for type I fCSA ($P = 0.077$, Fig. 1A). Total fCSA increased to a greater degree in the MIS group (MIS: 2449 ± 411 μm^2, COL: 1362 ± 262 μm^2; $P = 0.03$, ES = 0.88; Fig. 1C), likely driven by a greater increase in type II fCSA after 10 wk of RET (MIS: 2836 ± 478 μm^2, COL: 1546 ± 312 μm^2; $P < 0.05$, ES = 0.89; Fig. 1B).

Myonuclear and SC Response to Damage and Training

SC content. Basal SC content (total Pax7$^+$ cells/100 myofibers) increased significantly in both groups with training ($P < 0.05$). No differences were observed between groups in SC content in type I fibers (MIS: 4.69 ± 1.1 SC/fiber, COL: 3.28 ± 1.7 SC/fiber, $P > 0.05$), type II fibers (MIS: 8.74 ± 1.9 SC/fiber, COL: 6.33 ± 2.1 SC/fiber, $P > 0.05$), or when collapsed across fiber type (MIS: 13.4 ± 2.2 SC/fiber, COL: 9.60 ± 3.5 SC/fiber, $P > 0.05$; Table 1, Fig. 2).

Myonuclear accretion. The number of myonuclei per fiber increased significantly with training but was not different between groups in type I, type II, or total fibers at any time point ($P > 0.05$, Table 1). RET increased myonuclear number in type I fibers by ~14%, in type II fibers by ~26%, and in total fibers by ~20% (training effect, $P < 0.0001$), irrespective of supplement. The increase in myonuclei per fiber was positively correlated with fCSA in type I fibers in the COL group ($r = 0.68$, $P = 0.01$; Fig. 3A) but was not correlated with the increase in type I fCSA in the MIS group ($r = 0.12$, $P > 0.05$; Fig. 3D). Similarly, the change in myonuclei per fiber was positively correlated with fCSA in type II fibers in the COL group (MIS: $r = -0.19$, $P > 0.05$; COL: $r = 0.59$, $P = 0.03$; Fig. 3B) but was not correlated with the change in type II fCSA in the MIS group ($r = 0.19$, $P = 0.53$; Fig. 3E), and the same pattern followed for total fibers (MIS: $r = -0.02$, $P > 0.05$; COL: $r = 0.62$, $P = 0.02$; Figs. 3C, F).

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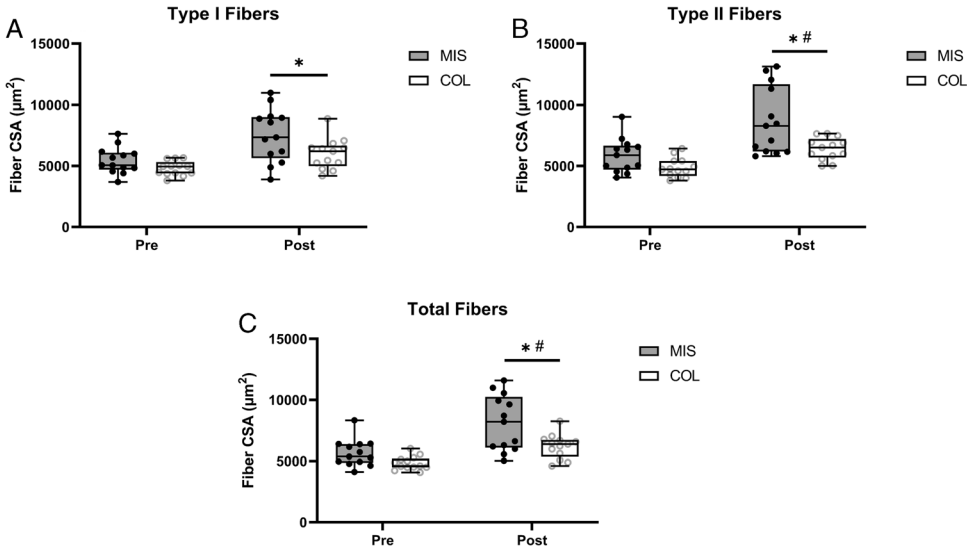

FIGURE 1—Type I (A), type II (B), and total (C) muscle fiber CSA for MIS (n = 13, shaded box) and COL (n = 13, open box) after 10 wk of RET. Values are presented as median lines and interquartile range (boxes) ± maximum and minimum values (whiskers), with + representing the mean. *Significantly different (P < 0.05) after training. #Significantly different (P < 0.05) between groups.

Myonuclear and SC domain. Myonuclear domain size increased with training but was not different between groups before or after 10 wk of RET (Table 1, Fig. 4). SC domain size was affected by neither supplementation (P = 0.86) or training (P = 0.19; Table 1, Figs. 5A–C).

SC content and activation in response to an acute bout of exercise—UTSR. The response to a single bout of exercise resulted in total Pax7$^+$ cells per 100 myofibers remaining unchanged at 48 h in both the MIS and COL groups across all fiber types (P > 0.05, Table 2). Pax7$^+$/MyoD$^+$ cells per 100 myofibers (indicating the presence of activated SC) were unchanged at 48 h in the MIS group in type I (0.01 ± 0.0 cells/100 myofibers), type II (1.16 ± 0.16 cells/100 myofibers), and total (1.17 ± 0.16 cells/100 myofibers) fibers after the single bout of damaging exercise as compared with Pre (type II: 0.65 ± 0.16 cells/100 myofibers; total: 0.66 ± 0.16 cells/100 myofibers; P < 0.05; Table 2, Figs. 6A–C). Pax7$^+$/MyoD$^-$ cells per 100 myofibers did not change from 48 h after the acute bout of damaging exercise (P > 0.05; Table 2, Figs. 6D–F).

SC content and activation in response to an acute bout of exercise—TSR. In response to an acute bout of damaging exercise of the same relative intensity following 10 wk of RET, total Pax7$^+$ cells/100 myofibers was not different for either group (MIS vs COL) or fiber type (type I, type II, total; P > 0.05, Table 2). Pax7$^+$/MyoD$^+$ cells per 100 myofibers were unchanged in both groups for type I and II fibers (Table 2; Figs. 6D, E), but there was a trend toward a significant interaction in total fiber SC activation in the trained state response (MIS: +2.71 ± 0.7 cells/100 myofibers, COL: +1.11 ± 0.5, P = 0.09, ES = 0.70; Table 2, Fig. 6F). Pax7$^+$/MyoD$^-$ cells per 100 myofibers did not change from Pre to 48 h after the acute bout of damaging exercise (P > 0.05, Table 2, Figs. 6A–C)

DISCUSSION

The purpose of this study was to assess the effect of a MIS on SC and muscle fiber properties following a 10-wk RET program in healthy young adults. We hypothesized that multiingredient supplementation would increase muscle fCSA, SC

TABLE 1. Fiber type–associated SC content (Pax7$^+$ cells/100 myofibers), myonuclei, SC domain, and myonuclear domain before and after 10 wk of RET, by condition.

		UT		TR	
		MIS	COL	MIS	COL
SC content (per 100 fibers)	I	9.34 ± 1.8	8.38 ± 1.47	14.03 ± 2.18*	11.66 ± 1.72*
	II	6.83 ± 0.9	6.56 ± 0.91	15.57 ± 2.33*	12.89 ± 2.02*
	Total	16.17 ± 2.4	14.94 ± 2.31	29.60 ± 3.58*	24.54 ± 3.47*
Myonuclei (per 100 fibers)	I	2.96 ± 0.14	3.00 ± 0.11	3.45 ± 0.20*	3.38 ± 0.16*
	II	2.81 ± 0.18	2.74 ± 0.10	3.55 ± 0.22*	3.45 ± 0.21*
	Total	2.88 ± 0.13	2.87 ± 0.09	3.50 ± 0.19*	3.42 ± 0.18*
SC domain (mm^2)	I	66.1 ± 23.8	33.7 ± 8.9	45.8 ± 9.9	40.3 ± 12.2
	II	53.7 ± 8.0	41.2 ± 11.6	45.6 ± 6.0	36.6 ± 8.8
	Total	49.5 ± 9.6	34.2 ± 9.2	41.0 ± 5.1	38.7 ± 11.4
Myonuclear domain (μm^2)	I	1790 ± 93	1614 ± 59	2035 ± 123*	1799 ± 102*
	II	1987 ± 130	1788 ± 82	2282 ± 150*	1908 ± 95*
	Total	1888 ± 90	1701 ± 58	2159 ± 128*	1854 ± 81*

Values are mean ± SE.
*Significant effect for exercise training (P < 0.05).
TR, trained; UT, untrained.

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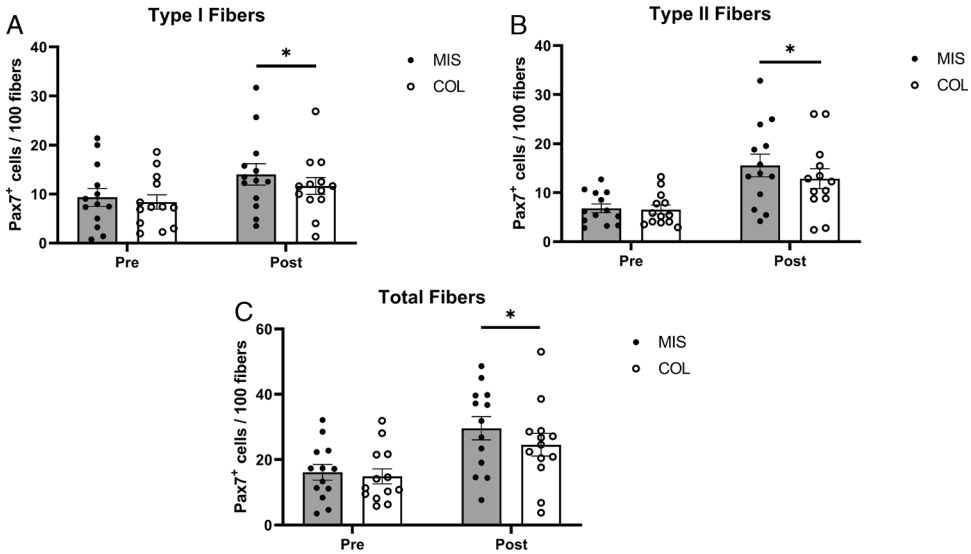

FIGURE 2—Mean number of SC per 100 fibers in type I (A), type II (B), and total (C) muscle fibers before (pre) and after (post) 10 wk of RET. *A significant effect of 10 wk training ($P < 0.05$).

content, myonuclear accretion, and myonuclear domain to a greater degree in MIS than COL. In this study, we demonstrate that individuals consuming the MIS exhibited greater increases in type II and total fCSA, but similar increases in type I fCSA. MIS also showed a trend toward a greater acute SC activation response in all fibers post-RET as compared with the COL group, with similar activation patterns in type I fibers. Myonuclear accretion increased significantly with training, with no differences between groups, but showed positive correlations to the change in fCSA only in the COL group. Myonuclear domain increased with training in both groups, and neither myonuclear nor SC domain was different between the MIS and COL groups.

Skeletal muscle hypertrophy in response to RET is characterized by increases in lean mass and strength, along with a significant increase in fCSA (32,33). In the present study, we observed greater increases in type II fCSA compared with type I fCSA (Fig. 1). Higher-intensity resistance exercise requires

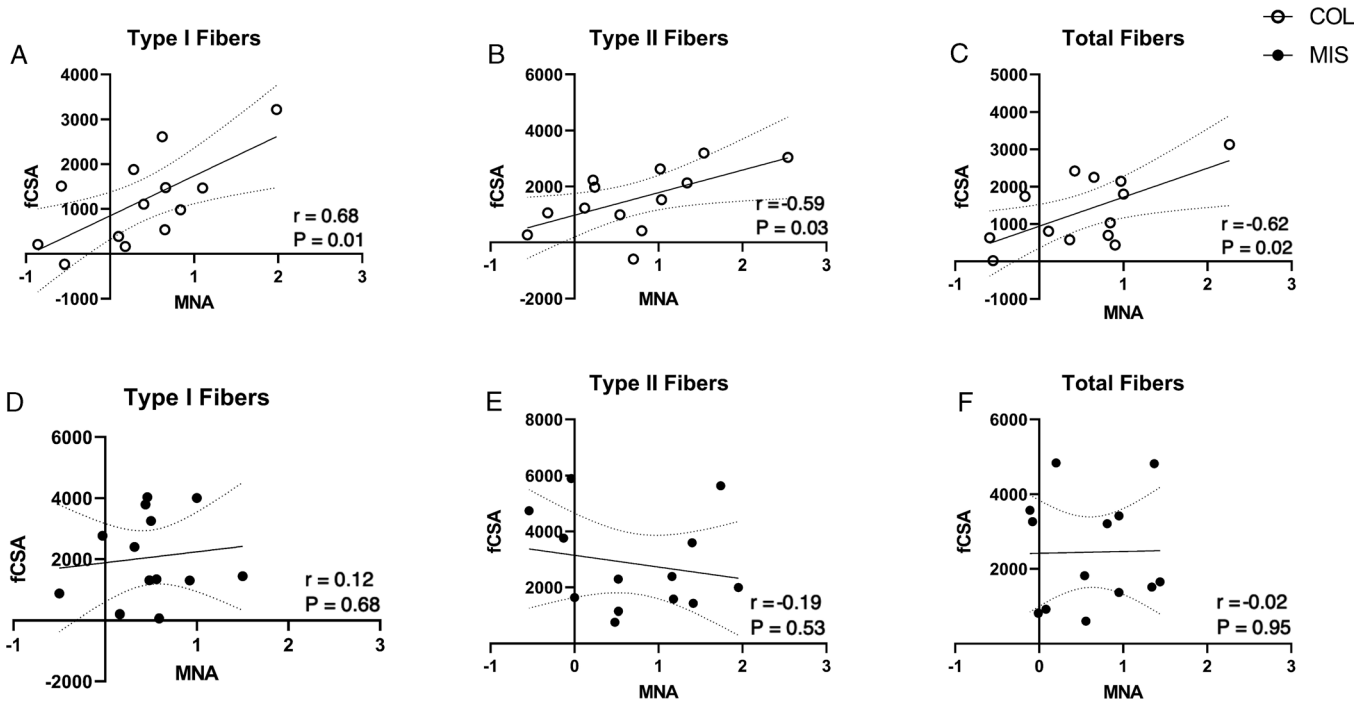

FIGURE 3—Correlation between type I (A, D), type II (B, E), and total (C, F) fCSA to MNA with 10 wk of RET. Open circles denote the COL group; closed circles denote the MIS group. *COL > MIS (supplement effect; $P < 0.05$). Data are presented as means with individual data points corresponding to a participant. MNA, myonuclear accretion.

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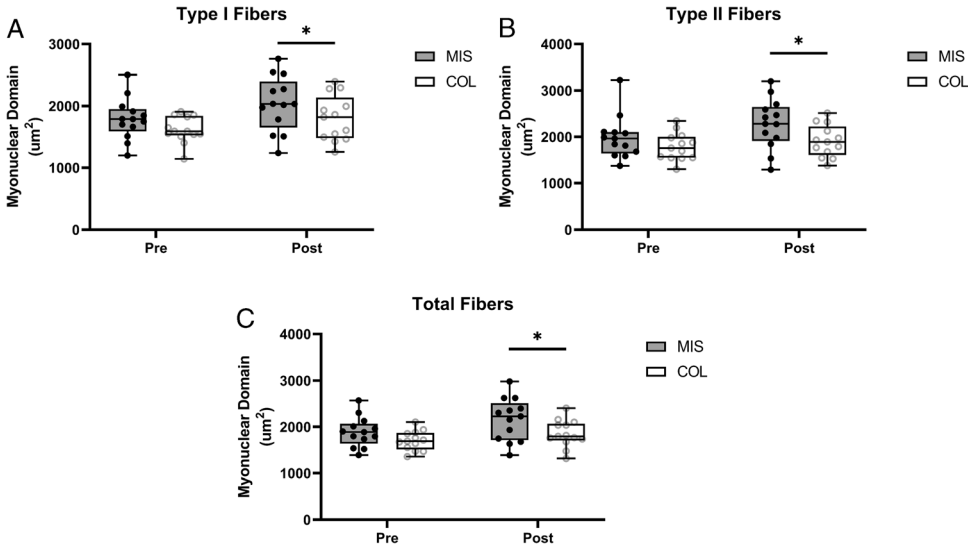

FIGURE 4—Myonuclear domain in type I (A), type II (B), and total (C) fibers before (Pre) and after (Post) 10 wk of RET. *Significantly different ($P < 0.05$) after training. #Significantly different ($P < 0.05$) between groups.

greater motor unit recruitment, ultimately engaging more type II fibers and leading to greater type II fiber growth at intensities >50% one-repetition maximum (1RM) (34–37). With MIS, we observed increases of 36% and 48% in type I and II fCSA, respectively, and with COL, we observed increases of 21% and 32% in type I and II fCSA, respectively. The increase in type I fCSA with MIS approached statistical significance ($P = 0.077$), whereas increases in type II fCSA were significantly greater with MIS ($P = 0.033$). The training protocol targeted approximately 70%–80% of the participants' 1RM in each exercise, and thus, although both fibers exhibited a significant response to training, type II fibers were more substantially affected by MIS. Given that the increase in type I fCSA was small (and nonsignificant), group-based differences (i.e., MIS vs COL) may have been undetectable. Previous literature combining protein supplementation (often whey) and RET (10–12 wk) have reported similar increases in type II fCSA, with a similar lack of detectable increase in type I fCSA (38).

Literature investigating whey protein, soy protein, or leucine alone or with the addition of creatine reported no significant benefit of supplementation in conjunction with an 8- to 12-wk RET program on increasing fCSA (38–41). Supplement studies investigating vitamin D have shown significant increases in fiber area with 10–12 wk of RET in healthy young adults (14,15,42,43) and, when combined together, may produce greater benefit as the established function of vitamin D is to promote calcium absorption to aid in the maintenance of blood calcium as well as bone mineralization (44). This points to possible synergistic effects of the two in augmenting the MIS formulation in the current study. Creatine alone has

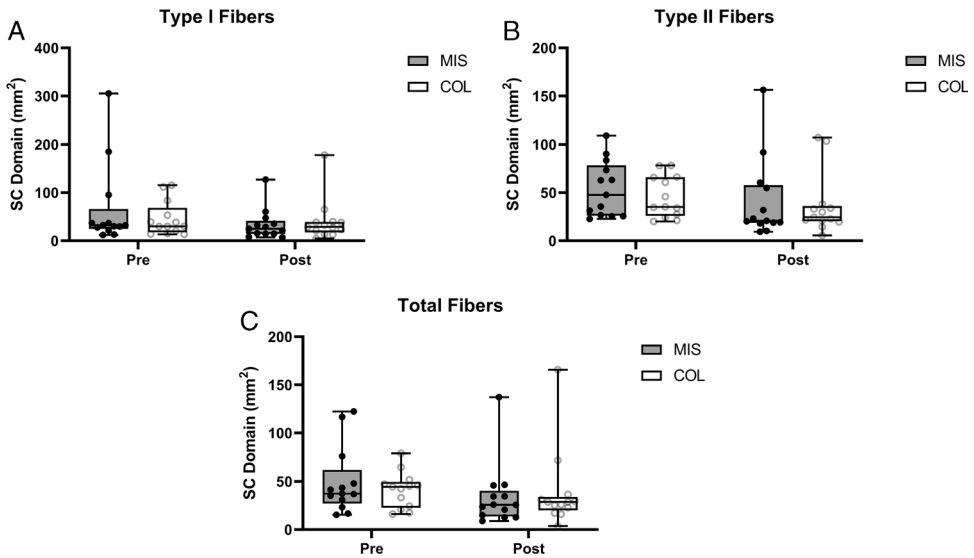

FIGURE 5—SC domain in type I (A), type II (B), and total (C) fibers before (Pre) and after (Post) 10 wk of RET. *Significantly different ($P < 0.05$) after training. #Significantly different ($P < 0.05$) between groups.

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TABLE 2. Fiber type–associated Pax7⁺ cells/100 myofibers, Pax7⁺/MyoD⁺ cells/100 myofibers, and Pax7⁺/MyoD⁻ cells/100 myofibers before and 48 h after 300 eccentric leg extensions in the untrained (UTSR) and trained (TSR) states by condition.

			UTSR		TSR	
		Fiber Type	Pre	Post	Pre	Post
Total Pax7⁺ cells per 100 myofibers	MIS	I	9.34 ± 1.82	10.48 ± 1.79	14.03 ± 2.18*	11.82 ± 1.76
		II	6.83 ± 0.90	9.47 ± 1.87	15.57 ± 2.33*	17.48 ± 2.61
		Total	16.17 ± 2.40	19.95 ± 3.53	29.60 ± 3.58*	28.14 ± 3.77
	COL	I	8.38 ± 1.47	8.56 ± 1.38	11.66 ± 1.72*	11.67 ± 2.08
		II	6.56 ± 0.91	7.55 ± 0.97	12.89 ± 2.02*	15.79 ± 2.30
		Total	14.94 ± 2.31	16.11 ± 2.12	24.54 ± 3.47*	24.70 ± 3.30
Pax7⁺/MyoD⁺ cells per 100 myofibers	MIS	I	0.01 ± 0.00	0.01 ± 0.00	0.02 ± 0.01	0.01 ± 0.01
		II	0.65 ± 0.16	1.16 ± 0.16	2.06 ± 0.35*	4.41 ± 0.59
		Total	0.66 ± 0.16	1.17 ± 0.16	1.73 ± 0.23*	4.43 ± 0.59
	COL	I	0.01 ± 0.00	0.02 ± 0.00	0.02 ± 0.00	0.03 ± 0.01
		II	0.96 ± 0.22	1.05 ± 0.17	2.41 ± 0.33*	3.64 ± 0.66
		Total	1.01 ± 0.21	1.01 ± 0.17	2.38 ± 0.33*	3.49 ± 0.63
Pax7⁺/MyoD⁻ cells per 100 myofibers	MIS	I	9.34 ± 1.82	10.47 ± 1.79	14.02 ± 2.18*	11.80 ± 1.76
		II	6.18 ± 0.99	8.31 ± 1.91	13.51 ± 2.15*	13.07 ± 2.81
		Total	15.52 ± 2.50	18.78 ± 3.54	27.87 ± 3.44*	23.71 ± 4.02
	COL	I	8.37 ± 1.48	8.54 ± 1.38	11.64 ± 1.73	11.64 ± 2.08
		II	5.60 ± 0.82	6.51 ± 0.94	10.48 ± 2.01*	12.15 ± 2.23
		Total	13.94 ± 2.21	15.11 ± 2.10	22.16 ± 3.46*	21.21 ± 3.09

Values are mean ± SE.
*Significant effect for exercise training ($P < 0.05$).

been shown to enhance increases in muscle mass and strength (45–48), as well as reduce leucine oxidation and urinary 3MH excretion (48,49), offering both an anabolic and anticatabolic benefit to users. With training, creatine has also shown greater enhancements in SC content and myonuclear concentration in skeletal muscle fibers (19). These benefits may be mediated by increases in myogenin and MRF4 mRNA and protein (50) as well as the upregulation of 4EBP1 and p70S6K phosphorylation (51,52) concomitant with a decrease in myostatin (53). Paired with our previous finding (9) that lean body mass increases were significantly greater in the MIS group, multi-ingredient supplementation may contribute to significantly greater increases in RET-induced muscle growth.

Increases in fCSA have been associated with the addition of myonuclei (3,20,32,33). According to the myonuclear domain theory (54), each myonucleus governs a particular volume of sarcoplasm over which it maintains transcriptional control (55). In the present study, there was significant myonuclear accretion over the 10 wk of RET. These increases reflect a main effect of time, but no interaction for condition. Despite a

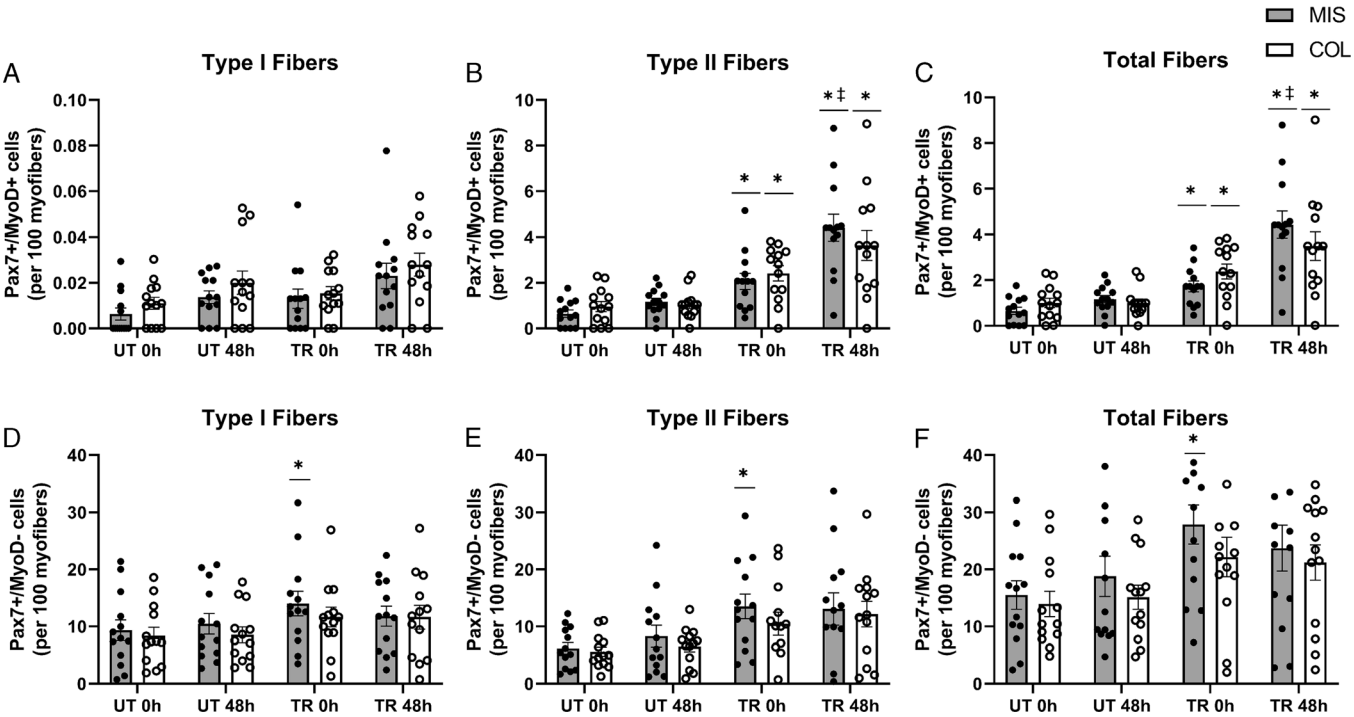

FIGURE 6—Quiescent (Pax7⁺/MyoD⁻; A–C) and activated (Pax7⁺/MyoD⁺; D–F) SC per 100 fibers in type I (A, D), type II (B, E), and total (C, F) muscle fibers before (0 h) and after (48 h) eccentric damage. *Significantly different ($P < 0.05$) with training status. #Significantly different ($P < 0.05$) between groups. ‡Significantly different after damage (48 h).

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greater increase in type II fCSA with MIS, there was no associated increase in myonuclear accretion or myonuclear domain. It has been postulated that a certain level of hypertrophy can be achieved before the addition of new nuclei are needed to support growth (54). It is possible that the upper limit of hypertrophy was not realized in each fiber type, and therefore, hypertrophy could occur to different degrees in each fiber type without the addition of new nuclei (54).

A curious finding from this study was the correlation between myonuclear accretion and the change in CSA in all fiber types in the COL group and not with MIS. The response observed by the COL group is in accordance with previous findings (56), where the change in myonuclear content before and after 8 wk of RET was positively associated with type I and type II muscle fiber hypertrophy. The MIS group, however, did not share the same associations between fCSA and myonuclear accretion. This finding suggests that muscle growth may be independent of the increase in myonuclear content when ingesting MIS. Whey protein and leucine in particular are known to activate mammalian target of rapamycin signaling, which phosphorylates eIF4E binding protein 1 and the ribosomal protein S6 kinase complex, both of which allow for increased translation of mRNAs of the protein synthesis pathway to occur (57). With MIS, large increases in fCSA were observed despite the absence of myonuclear addition, suggesting that MIS may support alternative signaling that can promote hypertrophy. For example, whey protein is able to target markers of ribosomal biogenesis, perhaps superseding the necessity of additional nuclei to contribute to muscle fiber growth (58). It is possible that muscle fiber hypertrophy is occurring through this mechanism without the contribution of new nuclei.

As noted previously, SC contributes to the growth of skeletal muscle with RET (59). RET stimulates an expansion of SC content, particularly associated with type II fibers (56,60,61). In the present study, we observed increases in SC content with training, particularly in type II fiber-associated SC, consistent with previous literature (56,62). Chronic overload has been shown to stimulate SC expansion associated with type II fibers, which may contribute to greater increases in type II fCSA, as seen in the current study. Interestingly, and along the same lines, we noticed a slightly greater increase in both type I and II fiber SC content expansion with the ingestion of the MIS, such that the MIS condition increased 50% and 190% in type I and II fiber SC content, respectively, whereas the COL condition increased 39% and 96%, respectively. Similarly, SC activation after a bout of damaging exercise increased robustly in both groups following 10 wk of RET. This is in accordance with findings reported by Nederveen et al. (63), where acute SC activation was more pronounced after a 16-wk progressive resistance training program, concomitant with an increase in MyoD gene expression in trained muscle (63). This robust increase in SC activation with RET is ostensibly an adaptive response to chronic exercise, and a function of increased capillarization and/or increased proximity of SC to capillaries (63). This allows for trained muscle to respond more efficiently to overload stimuli. Importantly, however, there was a trend for greater SC activation in the MIS versus COL group after training, particularly when collapsed across fiber type (P = 0.09, Figs. 6D–F). This may be an indicator that the combination of supplementation with training can induce more robust molecular responses after an acute bout of exercise regardless of fiber type. The lack of increase in $Pax7^{+}/MyoD^{-}$ cells per 100 fibers was as expected; no change in number has been seen in previous literature (63). Altogether, this indicates that the MIS may exert more potent effects than the COL supplement on SC activation in healthy young adults after training.

This study, to our knowledge, is the first to investigate the effect of two isoenergetic, isonitrogenous protein sources on myonuclear and SC dynamics in a population of healthy young men and women. Our comparison of MIS to an isonitrogenous, lower-quality collagen peptide-based supplement is, in our opinion, a more robust comparison than typical comparisons against a carbohydrate-based placebo. Collagen has been shown to be ineffective at stimulating muscle protein synthesis at rest or during exercise (64,65). As an incomplete protein source, specifically lacking the amino acid tryptophan, collagen peptides do not stimulate MPS to the same degree as whey protein (65). This is revealed in the COL supplement's lack of similar benefit compared with MIS. Thus, we present benefits of MIS consisting of whey protein, creatine, leucine, calcium citrate, and vitamin D on muscle mass and SC activation.

This study, as with most human physiological research, is not without its limitations. Many of the differences noted in this study were not of robust enough statistical significance to allow for definitive conclusions to be made, perhaps due to low sample size. Alternatively, the present results may inherently reflect the relative robustness of RET compared with supplementation on muscle mass. Morton et al. (12) report that supplementation only accounts for ~9% of gains in 1RM and ~27% of gains in fat-free mass. This suggests that the contribution of MIS to increases in fCSA, myonuclear accretion, and SC activation may be outweighed by RET in young men and women. Thus, although MIS may exert an effect in this population, this effect is more difficult to distinguish against the larger influence of RET, particularly without a non-protein placebo supplement or RET-only group.

We conclude that MIS supports augmented type II and total fCSA and the potential for greater SC activation post-RET in healthy young adults. In addition, we highlight the potential for larger increases in fiber size independent of myonuclear accretion with multi-ingredient protein supplementation. These results provide insight into the long-term effects of training on SC dynamics and fiber characteristics, demonstrating potential effects and benefits of ingesting whey-based multi-ingredient protein supplements as compared with collagen peptide–based supplements.

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